### Notations

Let $A$ and $B$ be two events.

(i) $A \cup B$ stands for the occurrence of $A$ or $B$ or both.

(ii) $A \cap B$ stands for the simultaneous occurrence of $A$ and $B$. $A \cap B$ can also be written as $AB$

(iii) $\overline{A}$ or $A'$ or $A^c$ stands for non-occurrence of $A$

(iv) $(A \cap \overline{B})$ stands for the occurrence of only $A$.

Priori : Knowledge which precedes from theoretical deduction or making assumption. Not from experience or observation

## 12.4 Probability

### 12.4.1 Classical definition (A priori) of probability (Bernoulli's principle of equally likely)

Earlier classes we have studied the frequency (A posteriori) definition of probability and the problems were solved. Now let us learn the fundamentals of the axiomatic approach to probability theory.

Posteriori : Knowledge which precedes from experience or observation.

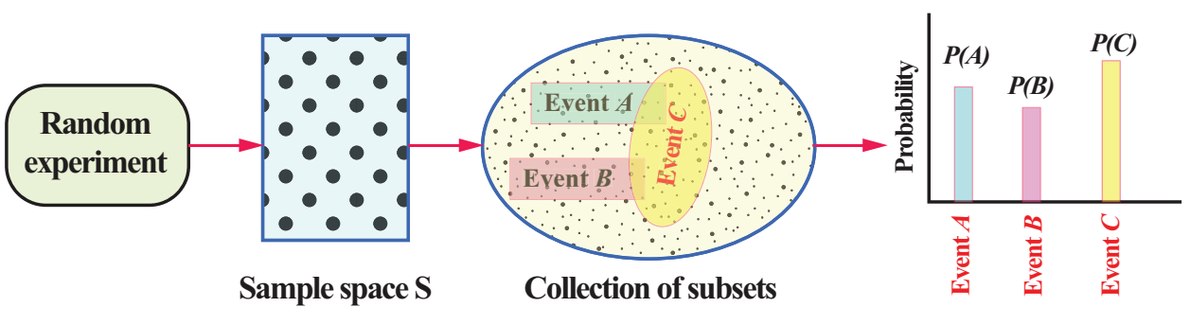
The basic assumption of underlying the classical theory is that the outcomes of a random experiment are equally likely. If there are $n$ exhaustive, mutually exclusive and equally likely outcomes of an experiment and $m$ of them are favorable to an event $A$, then the mathematical probability of $A$ is defined as the ratio $\frac{m}{n}$. In other words, $P(A) = \frac{m}{n}$.

**Definition 12.13**

Let $S$ be the sample space associated with a random experiment and $A$ be an event. Let $n(S)$ and $n(A)$ be the number of elements of $S$ and $A$ respectively. Then the probability of the event $A$ is defined as

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{Number of cases favourable to } A}{\text{Exhaustive number of cases in } S}$$

**Every probabilistic model involves an underlying process is shown in the following figure.**




The classical definition of probability is limited in its application only to situations where there are a finite number of possible outcomes. It mainly considered discrete events and its methods were mainly combinatorial. This renders it inapplicable to some important random experiments, such as 'tossing a coin until a head appears' which give rise to the possibility of infinite set of outcomes. Another limitation of the classical definition was the condition that each possible outcome is 'equally likely'.



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